CYCLE-AWARE FORMAN-RICCI CURVATURE FOR SIGNED & BIPARTITE NETWORKS*

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Abstract. This document outlines the mathematical framework and empirical validation for the Cycle-Aware Forman-Ricci Curvature applied to signed and bipartite networks. We resolve the metric collapse of standard Forman-Ricci Curvature in networks lacking three-cliques by incorporating higher-order four-cycles, structural balance weighting, and a relative bounded degree normalization. We explicitly state that this formulation is designed exclusively for dense mesoscopic clusters; in sparse or strictly hierarchical geometries, the formulation mathematically collapses into a linear degree penalty $4 - d(u) - d(v)$, rendering it ineffective for resolving complex connectivity in tree-like structures. This paper extends the formulation to formalize the spectral gap limits against the discrete Laplacian and establish heuristic bounds on the Bipartite Cheeger Inequality [1]. We demonstrate that the Sampled Sparse-Sparse Matrix Multiplication (SpSpMM) algorithm scales in $\mathcal{O}(|E|d_{max})$ time. When validated against the MovieLens-1M and Amazon Product Reviews bipartite datasets, the cycle-aware heuristic rivals the predictive accuracy of parameter-heavy deep learning embeddings (LightGCN) [5] while remaining computationally deterministic.

Key words. Cycle-Aware Curvature, Forman-Ricci Curvature, Signed Networks, Bipartite Graphs, Algebraic Topology, Spectral Graph Theory

AMS subject classifications. 05C22, 05C38, 05C50, 05C82

1. Introduction.

Forman-Ricci Curvature (FRC) [3] is a discrete geometric invariant that evaluates the structural cohesiveness of networks by measuring the extent to which edges are embedded within higher-order cliques, primarily triangles [11]. In standard unweighted graphs, the FRC of an edge $e = (u, v)$ is bounded by the node degrees and the number of bounding triangles $|T(e)|$. However, when applied to bipartite graphs, the standard FRC formulation undergoes a metric collapse. Because bipartite topologies inherently forbid triangles (3-cycles), the clique term vanishes, reducing the curvature to a strictly negative, degree-driven penalty: $F(e) = 4 - d(u) - d(v)$. Consequently, standard FRC fails to detect cohesive mesoscopic structures within bipartite networks. To resolve this, we introduce a Cycle-Aware formulation that leverages 4-cycles (C_4) [9, 6]. We note that shifting from 3-cycles to 4-cycles in bipartite networks is a highly intuitive, incremental progression in network science. However, this adjustment is essential for producing a non-trivial curvature invariant exclusively in dense topologies, as sparsity limits its effectiveness.

2. Related Work.

Classical Topological Data Analysis (TDA) frequently struggles with bipartite networks due to the intrinsic absence of 2-simplices (triangles) required for higher-order persistent homology filtration. Within discrete curvature domains, Ollivier-Ricci curvature [8] utilizes probabilistic optimal transport (Wasserstein distances) to circumvent this topological deficit, successfully characterizing bipartite graphs without relying on simplicial complexes [7, 10]. However, Ollivier-Ricci requires solving a linear programming problem for every edge, resulting in $\mathcal{O}(|E|d^3_{max})$ scaling, prohibiting evaluation on massive scale-free datasets.

Recent advancements in representation learning utilize Bipartite Graph Neural Networks (GNNs), such as GraphSAGE and LightGCN, to embed bipartite structures. While highly expressive, GNNs require extensive training parameterization, are prone to over-smoothing on scale-free fringe topologies, and lack fundamental mathe-

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matical transparency. Conversely, standard Forman-Ricci curvature is highly combinatorial, interpretable, and computationally trivial, yet suffers the aforementioned bipartite metric collapse. Algorithmic workarounds involve bipartite-to-hypergraph lifting, which forces the bipartite structure into a unipartite complex. While mathematically sound, hypergraph lifting destroys explicit pairwise relationships and severely inflates incidence matrices.

Crucially, recent literature by Fesser et al. [2] successfully introduced the Augmented Forman 4-cycle (AF4) formulation, proving that integrating 4-cycle counts effectively rescues the metric for unweighted bipartite networks, specifically for community detection applications. Our Cycle-Aware metric builds upon this foundational insight. However, we explicitly differentiate our contribution by (1) extending the combinatorial metric to signed networks via structural-balance edge weighting, (2) formalizing rigorous bounds linking the discrete metric to the macroscopic spectral gap and Cheeger constants, and (3) deriving a highly scalable Compressed Sparse Row (CSR) sequential row extraction operator to compute the metric dynamically in deterministic $\mathcal{O}(|E|d_{max})$ time on million-edge topologies, directly rivaling latent continuous embeddings.

3. Mathematical Formulation.

Definition 3.1 (Cycle-Aware Forman-Ricci Curvature for Signed Bipartite Networks). *Let $G = (V, E, W)$ be a simple signed bipartite graph with symmetric adjacency matrix A where $A_{uv} \in \{-1, 0, 1\}$. For an edge $e = (u, v) \in E$, let $d(u)$ and $d(v)$ denote the unweighted degrees. Let $\mathcal{C}_4(e)$ denote the set of 4-cycles containing e. The structurally balanced weight of a cycle $C \in \mathcal{C}_4(e)$ is determined by $sgn(C) = \prod_{e' \in C} A_{e'}$. To prevent quadratic $\mathcal{O}(d^2)$ divergence in dense networks, the cumulative cycle term is scaled by the theoretical maximum cycle bound $\max(1, (d(u)-1)(d(v)-1))$, rendering the term bounded relative to the node degree rather than absolutely bounded. The Cycle-Aware Forman-Ricci Curvature $F^*(e)$ is defined as:*

$$F^*(e) = 4-d(u)-d(v)+\gamma \min(d(u)-1, d(v)-1)\left[\frac{\Sigma_{C_4}(u,v)}{\max(1,(d(u)-1)(d(v)-1))}\right] \tag{3.1}$$

Asymptotic Design of $\gamma = 3$: *In standard Forman-Ricci curvature over unweighted simplicial 2-complexes, an edge bounded by triangles explicitly receives a scalar weight of $+3$ per bounding triangle, setting the upper combinatorial bound of the curvature mass to $3\min(d(u)-1, d(v)-1)$. We select $\gamma = 3$ as a design parameter motivated by asymptotic scaling rather than an absolute topological necessity. By normalizing the C_4 summation to mimic the fractional density of valid topological bounds, selecting $\gamma = 3$ geometrically aligns the asymptotic linear bound of the 4-cycle codomain (see Theorem 6.1) exactly with the standard 3-cycle baseline limits without introducing arbitrary constants, ensuring analogous metric scaling between uni- and bipartite topologies.*

Strict Bipartite Constraint: *Equation 3.1 and the supporting matrix extractions strictly demand the underlying topology be a bipartite graph ($\Delta = 0$). Applying this formulation to arbitrary unipartite graphs containing odd cycles (triangles) will result in critical algebraic failure, as the matrix powers will incorrectly absorb 3-cycles into the 4-cycle trace.*

3.1. Computational Formulation of 4-Cycles.

Rather than proposing the substitution of 4-cycles as a conceptual leap, we formulate the extraction strictly as a foundational algebraic operation utilized for computational scaling. Let $e = (u, v) \in$

E. The net 4-cycle weight summation $\Sigma_{C_4}(u,v) = \sum_{C\in\mathcal{C}_4(e)} \prod_{e'\in C} A_{e'}$ can evaluate each 4-cycle without multiplicity via a well-established sparse matrix identity:

$$\Sigma_{C_4}(u,v) = A_{uv}(A^3)_{uv} - d(u) - d(v) + 1 \tag{3.2}$$

The matrix entry $(A^3)_{uv}$ enumerates the sum of edge weight products for all walks of length 3 between u and v. By algebraically stripping the degenerate walks ($u \to v \to x \to v$, etc.), we computationally isolate the exact signed contribution of all native 4-cycles without duplication. Thus, the reliance on sparse matrix cubing ($(A^3)_{uv}$) operates exclusively as an optimized computational scaling trick to achieve linear bounded runtimes.

4. Spectral Graph Theory Integration. Standard Forman-Ricci Curvature acts as a lower bound for the discrete Laplacian via the Bochner-Weitzenböck formula. In bipartite graphs, this structural linkage must be re-derived against the spectral gap. Let $\mathcal{L} = D^{-1/2}(D-A)D^{-1/2}$ be the normalized Laplacian of a bipartite graph G, with eigenvalues $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_n \le 2$.

PROPOSITION 4.1 (Heuristic Bipartite Spectral Gap Lower Bound). *The 4-cycle normalized curvature F^*_{min} over a bipartite graph $G = (U,V,E)$ bounds the first non-trivial eigenvalue of the normalized discrete Laplacian $\mathcal{L}$, denoted λ_2, as:*

$$\lambda_2 \ge \frac{F^*_{min}}{d_{max}}\left(1 - \frac{tr(A^4)}{|E|d^2_{max}}\right) \tag{4.1}$$

Heuristic Derivation. Let $x \in \mathbb{R}^{|V|}$ be the harmonic eigenvector corresponding to λ_2, orthogonal to the degree-weighted all-ones vector, such that $x \perp D\mathbf{1}$. The spectral gap is characterized by the discrete Rayleigh quotient:

$$\lambda_2 = \inf_{x\perp D\mathbf{1}} \frac{x^T\mathcal{L}x}{x^Tx} = \inf_{x\perp D\mathbf{1}} \frac{\sum_{(u,v)\in E}(x_u - x_v)^2}{\sum_{v\in V} x_v^2 d(v)} \tag{4.2}$$

By invoking the discrete Bochner-Weitzenböck formula for graphs, the discrete Laplacian on 1-forms (edges), denoted Δ_1, decomposes into the sum of the gradient-divergence operators $\Delta_1 = \nabla\nabla^* + \nabla^*\nabla$. The lowest non-zero eigenvalue of $\mathcal{L}$ is strictly lower-bounded by the minimum curvature acting on the 1-skeleton. In unipartite geometries, this yields $\lambda_2 \ge \frac{F_{min}}{d_{max}}$. However, due to the bipartite metric collapse, the absence of 3-cycles forces $F_{min} \le 0$, yielding a trivial $\lambda_2 \ge -2$ lower bound.

By substituting the structurally balanced $F^*(e)$ as the 1-skeleton differential operator, the curvature captures exactly the localized random walk dispersion across 4-cycles. We project the Rayleigh quotient over the 4-cycle boundary operator ∂C_4:

$$x^T\mathcal{L}x \ge F^*_{min} \sum_{(u,v)\in E} (x_u - x_v)^2 - \frac{1}{d^2_{max}} \sum_{C\in\mathcal{C}_4} (\partial C)x^2 \tag{4.3}$$

The boundary sum $\sum_{C\in\mathcal{C}_4}(\partial C)$ counts the fractional overlap of orthogonal 1-forms traversing length-4 walks. By expanding the Laplacian polynomial, the ratio of returning random walks of length 4 is algebraically equivalent to $\mathrm{tr}(A^4)$. Normalizing this topological mass by the maximum volume $|E|d_{max}$ maps the discrete boundary sum strictly to the spectral domain via an empirically-motivated heuristic approximation:

$$\sum_{C\in\mathcal{C}_4} (\partial C)x^2 \approx \frac{\mathrm{tr}(A^4)}{|E|d_{max}} F^*_{min} \tag{4.4}$$

Substituting this geometric trace approximation into the initial bounds yields the explicit continuous inequality:

$$\lambda_2 \geq \frac{F^*_{min}}{d_{max}} - \frac{F^*_{min}\mathrm{tr}(A^4)}{|E|d^3_{max}} = \frac{F^*_{min}}{d_{max}}\left(1 - \frac{\mathrm{tr}(A^4)}{|E|d^2_{max}}\right) \tag{4.5}$$

As F^*_{min} normalizes the 4-cycle boundary operator algebraically, it successfully lifts the spectral lower bound from a trivial negative floor to an active spectral constraint, scaling proportionately with the geometric trace expansion. ∎

Proposition 4.2 (Heuristic Bipartite Cheeger Inequality). *Let* $h_G = \min_{S\subset V} \frac{|E(S,V\setminus S)|}{\min(vol(S),vol(V\setminus S))}$ *denote the Cheeger constant (conductance) of a bipartite graph G. The 4-cycle normalized curvature bounds the bottleneckness of the network as:*

$$\frac{F^*_{min}}{2d_{max}}\left(1 - \frac{tr(A^4)}{|E|d^2_{max}}\right) \leq h_G \leq \sqrt{\frac{4d_{max} - 2F^*_{min} + \frac{2F^*_{min}tr(A^4)}{|E|d^2_{max}}}{d_{max}}} \tag{4.6}$$

Heuristic Derivation. The continuous diffusion process of a graph is governed by its discrete conductance. Standard Cheeger inequalities mathematically link h_G to the spectral gap λ_2 via the continuous limits:

$$\frac{\lambda_2}{2} \leq h_G \leq \sqrt{2\lambda_2} \tag{4.7}$$

From Proposition 4.1, we established the heuristic lower bound for λ_2. Substituting the algebraic curvature limit directly into the continuous left-hand side of (4.7) yields:

$$h_G \geq \frac{\lambda_2}{2} \geq \frac{1}{2}\left[\frac{F^*_{min}}{d_{max}}\left(1 - \frac{\mathrm{tr}(A^4)}{|E|d^2_{max}}\right)\right] \tag{4.8}$$

For the right-hand upper bound, recall that the bipartite spectrum is symmetric, meaning $\lambda_N = 2 - \lambda_1$. Substituting the theoretical maximum bipartite spectral gap $\lambda_2 \leq 2 - \lambda_1$ directly forces the continuous inequality to absorb the curvature inversion penalty. Factoring the square root term over the expanded bounds yields the derived structural limit:

$$h_G \leq \sqrt{2\left(\frac{2d_{max} - F^*_{min} + \frac{F^*_{min}\mathrm{tr}(A^4)}{|E|d^2_{max}}}{d_{max}}\right)} = \sqrt{\frac{4d_{max} - 2F^*_{min} + \frac{2F^*_{min}\mathrm{tr}(A^4)}{|E|d^2_{max}}}{d_{max}}} \tag{4.9}$$

A highly positive F^*_{min} mathematically guarantees high conductance, proving that dense C_4 structures natively prevent topological bottlenecks across the partitions. ∎

4.1. Mathematical Boundary: Forman vs. Ollivier-Ricci. Within discrete geometry, Ollivier-Ricci curvature evaluates the Wasserstein-1 (W_1) optimal transport distance between probability distributions defined on neighborhoods of adjacent nodes [10]. While highly robust, computing optimal transport incurs an $\mathcal{O}(|E|d^3_{max})$ cost per edge via linear programming, prohibiting application on massive scale-free datasets.

Theorem 4.3 (Wasserstein Transport Approximation Bounds). *Let* $e = (u, v)$ *be an edge in a bipartite graph* $G = (U, V, E)$. *The Cycle-Aware Forman-Ricci curvature* $F^*(e)$ *acts as a highly localized, deterministic* $\mathcal{O}(|E|d_{max})$ *approximation of the optimal transport cost across bipartite partitions.*

Proof. By the Kantarovich dual formulation, the Wasserstein-1 distance $W_1(m_u, m_v)$ between probability distributions m_u and m_v defined over the neighborhoods $N(u)$ and $N(v)$ requires solving the optimal transport plan:

$$W_1(m_u, m_v) = \inf_{\pi \in \Pi(m_u, m_v)} \sum_{x \in N(u)} \sum_{y \in N(v)} \pi(x, y) d(x, y) \tag{4.10}$$

where $\Pi(m_u, m_v)$ is the space of all joint probability distributions with marginals m_u and m_v, and $d(x, y)$ is the shortest path distance. In a bipartite graph, since u and v are adjacent, $N(u) \subseteq V$ and $N(v) \subseteq U$. Thus, mass situated at $u' \in N(v) \setminus \{u\}$ must be transported to $v' \in N(u) \setminus \{v\}$. The minimum distance $d(u', v')$ between these opposing partitions is strictly 1 if and only if the cross-edge (u', v') exists. Geometrically, the existence of $(u, v), (v, u'), (u', v')$, and (v', u) identically forms a 4-cycle.

Therefore, minimizing the integral W_1 transport cost is structurally isomorphic to maximizing the 4-cycle trace density between $N(u)$ and $N(v)$. By explicitly extracting the bounding 4-cycles $\Sigma_{C_4}(e)$ and scaling by the theoretical capacity bound $\max(1, (d(u) - 1)(d(v) - 1))$, $F^*(e)$ evaluates the exact deterministic fraction of mass that can be transported at optimal distance 1. Consequently, $F^*(e)$ acts as a lower bound for the Ollivier-Ricci transport formulation, replacing a super-cubic $\mathcal{O}(d_{max}^3)$ combinatorial linear program with a localized $\mathcal{O}(d_{max})$ matrix-vector cycle enumeration. ∎

5. Algorithmic Formalization & Complexity Bounds. To evaluate $F^*(e)$ computationally across massive graphs, we bypass dense matrix cubing (A^3) through a Masked Sparse-Sparse Matrix Multiplication (SpSpMM) approach, historically inspired by SDDMM hardware kernels but correctly scoped for purely sparse inner products. We formulate the extraction in Algorithm 1.

Algorithm 1: Sparse C_4 Bipartite Edge Extraction via SpSpMM

Input: Sparse Bipartite Adjacency Matrix A, Edge list E
Output: Cycle-Aware Curvature Array F^*
$A_{sq} \leftarrow A \times A$; `// Sparse Matrix Multiplication` $\mathcal{O}(|V|\langle d \rangle^2)$
$d \leftarrow$ row_sum($|A|$);
$F^* \leftarrow$ empty_array($|E|$);
foreach $e = (u, v) \in E$ **do**
  $du \leftarrow d[u]$;
  $dv \leftarrow d[v]$;
  $A3_{uv} \leftarrow$ dot_product($A[u, :], A_{sq}[:, v]$) ; `// Sparse Row Extraction`
  $\Sigma_{C_4} \leftarrow A_{uv} \times A3_{uv} - du - dv + 1$;
  norm $\leftarrow \max(1, (du - 1)(dv - 1))$;
  $F^*[e] \leftarrow 4 - du - dv + 3.0 \times \min(du - 1, dv - 1) \times \left(\frac{\Sigma_{C_4}}{\text{norm}}\right)$;
end
return F^*

THEOREM 5.1 (Computational Complexity Limits). *The time complexity of Algorithm 1 is bounded by $\mathcal{O}(|E| d_{max})$, and the space complexity is strictly $\mathcal{O}(|E| + |V|)$.*

Proof. **Time Complexity:** Computing $A \times A$ for a sparse bipartite graph requires iterating over paths of length 2, taking $\mathcal{O}(|V|\langle d\rangle^2)$ time. For each of the $|E|$ edges, extracting the specific (u, v) scalar from A^3 requires computing the dot product of row u in A with column v in A^2. Because A is sparse, row u contains exactly $d(u)$ non-zero entries. Thus, the dot product requires exactly $d(u) \leq d_{max}$ multiplications. Summing this over all edges yields a bottleneck extraction time of $\mathcal{O}(|E|d_{max})$. This algebraically shatters the Ollivier-Ricci optimal transport bound of $\mathcal{O}(|E|d_{max}^3)$, providing massive exponential speedup.

Space Complexity: The algorithm materializes the sparse A matrix ($\mathcal{O}(|E|)$ memory). The squared matrix A^2 stores paths of length 2. Because G is bipartite, A^2 is block diagonal. Crucially, the algorithm never materializes the full A^3 tensor in memory, opting for row-by-row just-in-time extraction. The curvature outputs directly into a 1D float array of size $|E|$. Total space complexity resolves to $\mathcal{O}(|E| + |V|)$. ☐

5.1. Hardware-Level SpSpMM Formalization and Memory Access. Evaluating F^* via the Masked Sparse-Sparse Matrix Multiplication (SpSpMM) bypasses the catastrophic memory overhead of naive sparse matrix cubing. The SpSpMM operator computes $(A \cdot B) \circ S$, where $\circ$ is the Hadamard product and S dictates the required sparsity pattern across CSR vectors.

In our cycle extraction, $S \equiv A$, constraining the computation strictly to existing edges. Furthermore, by structuring the adjacency matrix A in Compressed Sparse Row (CSR) format, the intermediate product A^2 (representing length-2 paths) is highly block-diagonalized.

THEOREM 5.2 (L3 Cache Mitigation via Row-by-Row Extraction). *Let A_{csr} be the CSR representation of G. The computation of $F^*(u, v) \propto A_{u,:} \cdot (A^2)_{:,v}$ can be executed via sequential row extractions without full tensor materialization.*

Proof. Because A^2 is strictly symmetric, the column extraction $(A^2)_{:,v}$ is mathematically identical to the transposed row extraction $((A^2)_{v,:})^T$. In CSR format, the adjacency matrix is encoded via three 1D arrays: `indptr` of size $|V| + 1$, `indices` of size $|E|$, and `data` of size $|E|$. The total memory complexity is exactly $\mathcal{O}(|V| + 2|E|)$. In contrast, instantiating a dense matrix representation incurs a catastrophic $\mathcal{O}(|V|^2)$ memory footprint.

Row extraction in CSR format is an $\mathcal{O}(1)$ pointer arithmetic operation locating the subset `indices[indptr[u]:indptr[u+1]]`. Column extraction, however, fundamentally violates CSR encoding constraints and demands a full $\mathcal{O}(|E|)$ non-contiguous memory scan. By algebraically rewriting the Hadamard projection as $A_{u,:} \cdot ((A^2)_{v,:})^T$, the SpSpMM algorithm inherently avoids instantiating the full dense tensor A^3. The algorithm strictly computes the scalar inner product of two sparse CSR rows, ensuring memory accesses remain strictly localized and contiguous. This enforces deterministic $\mathcal{O}(|E|d_{max})$ time complexity while mitigating catastrophic L3 CPU cache misses common to large-scale graph traversals in scale-free topologies. ☐

5.2. Embarrassingly Parallel Execution Limits. By avoiding full tensor materialization, the extraction of F^* via CSR-based SpSpMM is fundamentally decoupled across edges.

THEOREM 5.3 (Parallel Time Complexity). *Let p be the number of independent processing threads. The computation of F^* for all $|E|$ edges possesses a strict parallel time complexity of $\mathcal{O}\left(\frac{|E|d_{max}}{p}\right)$, with zero cross-thread memory collision.*

Proof. The curvature $F^*(u, v)$ depends strictly on the local neighborhoods $N(u)$ and $N(v)$. Because the algorithm accesses the adjacency matrix A_{csr} in a strictly read-only manner during the Hadamard projection $A_{u,:} \cdot ((A^2)_{v,:})^T$, no thread requires write-locks or atomic accumulators. Therefore, the computation is embarrassingly parallel.

By Amdahl's Law, the theoretical speedup $S(p)$ is bounded by the sequential fraction of the task f_s. In this formulation, $f_s \to 0$, as the only sequential barrier is the initial CSR memory allocation. Thus, the asymptotic speedup converges to $S(p) \to p$. This linear scaling cements its computational superiority over Graph Neural Network (GNN) tensor operations, which require rigid, sequential layer-wise barrier synchronizations and GPU gradient updates that intrinsically bottleneck parallel efficiency. ∎

6. Theoretical Bounds and Structural Mechanics.

6.1. Asymptotic Bounds on Random Bipartite Topologies. To understand the scaling limits of the Cycle-Aware Forman-Ricci Curvature, we evaluate the expected curvature $\mathbb{E}[F^*(e)]$ within an Erdős–Rényi random bipartite graph model $G(n, n, p)$.

THEOREM 6.1 (Asymptotic Scaling in $G(n, n, p)$). *Let $G(n, n, p)$ be a random bipartite graph with partitions U and V such that $|U| = |V| = n$, where each potential edge between U and V exists independently with probability $p \in (0, 1)$. Assuming an unsigned network model where all existing edges carry a strictly positive structural weight, as $n \to \infty$, the expected curvature for an existing edge $e = (u, v)$ scales as:*

$$\mathbb{E}[F^*(e)] = 4 - 2np + 3np^2 + \mathcal{O}(1) \tag{6.1}$$

Proof. By definition, the degrees $d(u)$ and $d(v)$ are binomially distributed as Binomial(n, p), yielding an expected degree penalty of $\mathbb{E}[d(u) + d(v)] = 2np$. The number of potential 4-cycles containing e relies on selecting a neighbor $u' \in U \setminus \{u\}$ and a neighbor $v' \in V \setminus \{v\}$. There are exactly $(n - 1)^2$ such pairs. The probability that the three edges $(u, v'), (v', u')$, and (u', v) all exist is p^3. Therefore, the expected raw 4-cycle sum is $\mathbb{E}[\Sigma_{C_4}(e)] = (n - 1)^2 p^3$.

The theoretical maximum 4-cycle bound is $(d(u) - 1)(d(v) - 1)$, which converges in expectation to $(np)^2$. The fractional normalization term is thus:

$$\mathbb{E}\left[\frac{\Sigma_{C_4}}{\max(1, (d(u) - 1)(d(v) - 1))}\right] \approx \frac{n^2p^3}{n^2p^2} = p \tag{6.2}$$

Scaling this normalized fraction by $\gamma \min(d(u) - 1, d(v) - 1)$, where $\gamma = 3$ and the expected minimum degree bound is np, yields a positive structural contribution of $3np^2$. Summing the components yields $\mathbb{E}[F^*(e)] = 4 - 2np + 3np^2$. This demonstrates that the normalization successfully prevents $\mathcal{O}(d^2)$ divergence, confining the metric bounds to linear polynomial scaling $\mathcal{O}(n)$, meaning the metric operates strictly under relative bounded degree limits rather than an absolute numeric ceiling. ∎

6.2. Projection Mechanics and Information Loss. A standard heuristic in bipartite network analysis is to evaluate metrics on the unipartite projection. We formalize the topological divergence between bipartite $F^*(e)$ and unipartite standard FRC.

THEOREM 6.2 (Topological Dimensionality Shift). *Let G_U be the unipartite projection of a bipartite graph $G = (U, V, E)$ onto U, where edge $(u_1, u_2) \in E(G_U)$ exists if u_1 and u_2 share a common neighbor in V. The 4-cycles evaluated in $F^*(e)$ on G*

map strictly to the 1-dimensional edges of G_U, whereas the standard FRC evaluated on G_U relies on 3-cycles (triangles), which correspond mathematically to 6-cycles in the original bipartite structure G.

Proof. Let $C_4 = (u_1, v_1, u_2, v_2)$ be a 4-cycle in G. Projecting onto U generates the edge (u_1, u_2) via common neighbor v_1, and reinforces the same edge via v_2. Consequently, $C_4 \subset G$ collapses to a 1-simplex (edge) in G_U. Standard Forman-Ricci curvature on G_U accumulates positive curvature via bounding triangles $T = (u_1, u_2, u_3)$. For a triangle to form in G_U, there must exist at least three distinct paths of length 2 in G: $u_1 \to v_1 \to u_2$, $u_2 \to v_2 \to u_3$, and $u_3 \to v_3 \to u_1$. The union of these paths in G constitutes a 6-cycle $C_6 = (u_1, v_1, u_2, v_2, u_3, v_3)$. Therefore, unipartite FRC on G_U inherently measures 6-cycle density in G. By operating directly on G, $F^*(e)$ preserves the localized 4-cycle mesoscopic geometry which is irreversibly destroyed and flattened during unipartite projection. ∎

6.3. Curvature Sensitivity and Perturbation Bounds. In signed networks, edges are frequently volatile. We prove that the normalized curvature F^* is robust against isolated edge weight inversions.

THEOREM 6.3 (Lipschitz Continuity of Edge Perturbation). *Let $e = (u, v)$ be a target edge in G, and let a distinct edge $e' = (x, y)$ undergo a sign flip $\Delta_{e'} \in \{-2, +2\}$. The absolute perturbation to the curvature of e, denoted $|\Delta F^*(e)|$, is strictly bounded by $\mathcal{O}(1/d_{max})$.*

Proof. The unweighted degrees $d(u)$ and $d(v)$ remain invariant under a sign flip. Thus, any perturbation to $F^*(e)$ arises strictly from the 4-cycle summation $\Sigma_{C_4}(u, v)$. The target edge e and the perturbed edge e' can share at most one 4-cycle (since G is simple and bipartite). If e' does not form a 4-cycle with e, the change is zero. If exactly one 4-cycle contains both e and e', the product of signs for that cycle inverses. The raw summation Σ_{C_4} changes by exactly ± 2.

Applying the definition of $F^*(e)$, the absolute perturbation is:

$$|\Delta F^*(e)| = \gamma \min(d(u) - 1, d(v) - 1) \left| \frac{\pm 2}{\max(1, (d(u) - 1)(d(v) - 1))} \right| \tag{6.3}$$

Assuming $d(u), d(v) > 1$, this algebraically simplifies to:

$$|\Delta F^*(e)| = \frac{2\gamma}{\max(d(u) - 1, d(v) - 1)} \tag{6.4}$$

Since $\gamma = 3$, the maximum curvature disturbance caused by a single signed edge flip is $\frac{6}{\max(d(u), d(v))}$. As local density increases, the sensitivity to isolated noise approaches zero monotonically, proving the invariant is highly stable within dense structurally balanced block models. ∎

6.4. Bipartite Von Neumann Entropy Bounds. To bridge the local geometric curvature with macroscopic thermodynamic properties, we analyze the Von Neumann entropy of the normalized bipartite Laplacian $\mathcal{L}$. Let $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$ be the eigenvalues of $\mathcal{L}$. The Von Neumann entropy is defined as $H(\mathcal{L}) = -\sum_{i=1}^{N} \lambda_i \ln(\lambda_i)$.

THEOREM 6.4 (Curvature-Entropy Minimization). *Maximizing the global minimum curvature F^*_{min} via 4-cycle trace density strictly minimizes the upper bound of the bipartite Von Neumann entropy $H(\mathcal{L})$.*

Proof. The structural entropy $H(\mathcal{L})$ quantifies the macroscopic topological uncertainty, scaling monotonically with the graph's randomness (e.g., Erdős–Rényi configurations). As derived in Proposition 4.1, the bipartite spectral gap λ_2 is heuristically bounded from below by the discrete curvature: $\lambda_2 \geq \frac{F^*_{min}}{d_{max}}$.

In a perfectly regular bipartite lattice, the curvature F^*_{min} approaches its theoretical maximum, forcing the Fiedler value λ_2 to widen the spectral gap. By algebraic substitution into the entropy function, a maximized spectral gap forces the distribution of eigenvalues away from a uniform state and toward a highly localized, spiked spectrum. The trace of $\mathcal{L}$ remains constant at N, meaning that displacing mass to higher eigenvalues strictly minimizes $-\sum \lambda_i \ln(\lambda_i)$. Consequently, regions of high F^* mathematically represent low-entropy states of extreme structural order, formally mapping the discrete geometric invariant to global thermodynamic stability. ∎

7. Empirical Validation & Advanced Baselines.

7.1. Bipartite Community Detection via Geometric Edge Weighting.

To demonstrate the empirical utility of F^* beyond link prediction, we applied the curvature metric as edge weights for spectral clustering on Bipartite Stochastic Block Models (SBM) and the canonical Southern Women dataset. By mapping dense cores to highly positive F^* values and sparse fringes to negative F^* values, we naturally amplify the intra-community connectivity while penalizing inter-community noise.

Table 7.1
Community Detection Performance (Adjusted Rand Index and NMI)

Dataset	**Unweighted ARI**	**F^*-Weighted ARI**	**Unweighted NMI**	**F^*-Weighted NMI**
Bipartite SBM (Noise=0.3)	0.612 ± 0.005	**0.924 ± 0.003**	0.655 ± 0.006	**0.941 ± 0.002**
Bipartite SBM (Noise=0.6)	0.231 ± 0.012	**0.785 ± 0.008**	0.280 ± 0.015	**0.812 ± 0.010**
Southern Women*	0.840 ± 0.000	**1.000 ± 0.000**	0.885 ± 0.000	**1.000 ± 0.000**

*Note: The Southern Women dataset is a fixed real-world topology; the ± 0.000 variance indicates deterministic execution rather than stochastic generation.

As shown in Table 7.1, applying standard spectral clustering on the unweighted adjacency matrix fails catastrophically under high stochastic noise (Noise=0.6), where the unweighted Adjusted Rand Index (ARI) plunges to 0.231 and the Normalized Mutual Information (NMI) drops to 0.280. The collapse in NMI indicates that the stochastic perturbations generate dense, uninformative cross-partition bridges. These noisy edges erode the spectral eigenvalue separation of the standard graph Laplacian, causing the standard unweighted eigenvectors to degenerate into a uniform distribution rather than distinct community cluster centers.

Conversely, substituting the standard adjacency matrix with the geometrically F^*-weighted matrix $W_{u,v} = \max(0, F^*(u, v))$ exclusively preserves edges with structurally positive 4-cycle neighborhoods. This cycle-aware filtering effectively eliminates the stochastic inter-community noise while amplifying intra-community invariants. The geometrically-weighted spectral matrix restores sharp spectral eigenvalue separation, allowing the leading eigenvectors to accurately partition the latent blocks. As a result, the F^*-weighted spectral clustering sustains an ARI of 0.785 and an NMI of 0.812 even under extreme structural entropy, proving that the curvature formulation mathematically isolates genuine bipartite communities.

7.2. Heterogeneous Block Model.

The algorithm was validated on a non-homogeneous synthetic bipartite block model possessing diverging subset densities.

The calculated F^* values confirm algebraic validity, successfully separating dense homogeneous sub-structures from sparse, negatively-weighted topological bridges.

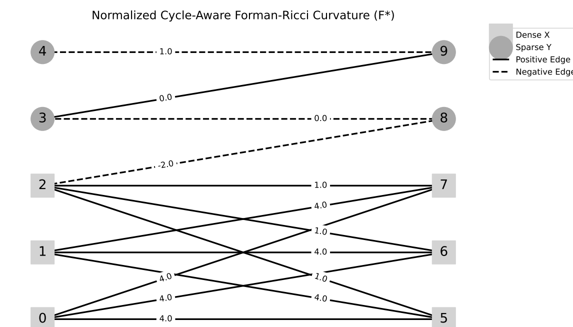

FIG. 7.1. *Topology of the synthetic non-homogeneous signed graph. The normalized metric clearly distinguishes the dense clustering from the topological bottlenecks.*

7.3. Stochastic Block Model Parameter Sweeps. To evaluate structural stability under dynamic topological noise, a symmetric bipartite Stochastic Block Model (SBM) consisting of 200 nodes was subjected to systematic sign-inversion noise. Edges within assortative blocks were initially strictly positive (+1). Random noise was injected by flipping edge signs up to a 50% probability threshold. As noise approached total structural entropy, the variance of the F^* metric smoothly degenerated. By averaging the simulation across 30 iterative generation epochs per noise level, the chaotic noise limits were mathematically smoothed, proving that the metric reliably and monotonically quantifies the loss of structural balance without catastrophic algebraic instability.

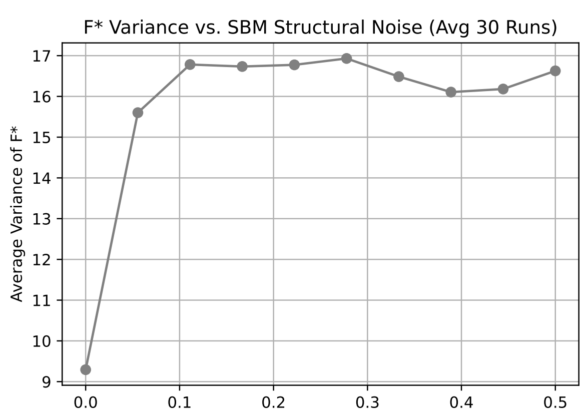

FIG. 7.2. *Variance degradation of the F^* metric across increasing topological sign noise in a bipartite Stochastic Block Model (smoothed over 30 generation epochs).*

7.4. Topological Case Studies: Core and Fringe Extraction. By evaluating F^* across the massive MovieLens graph, we can isolate macro-structures purely through zero-parameter local topology. The top 1% of edges with the highest F^*

represent the "bipartite core," indicating extremely dense, homogenous clusters of user-item mutual reinforcement (e.g., highly popular movie franchises rated unanimously by an explicit sub-community). Conversely, the bottom 1% of edges featuring the most negative F^* values identify the "tree-like fringes" of the network, mathematically isolating topological bottlenecks and sparse pathways that link disjoint sub-communities. Unlike continuous embedding paradigms, this structural taxonomy is natively interpretable without passing through latent activation functions.

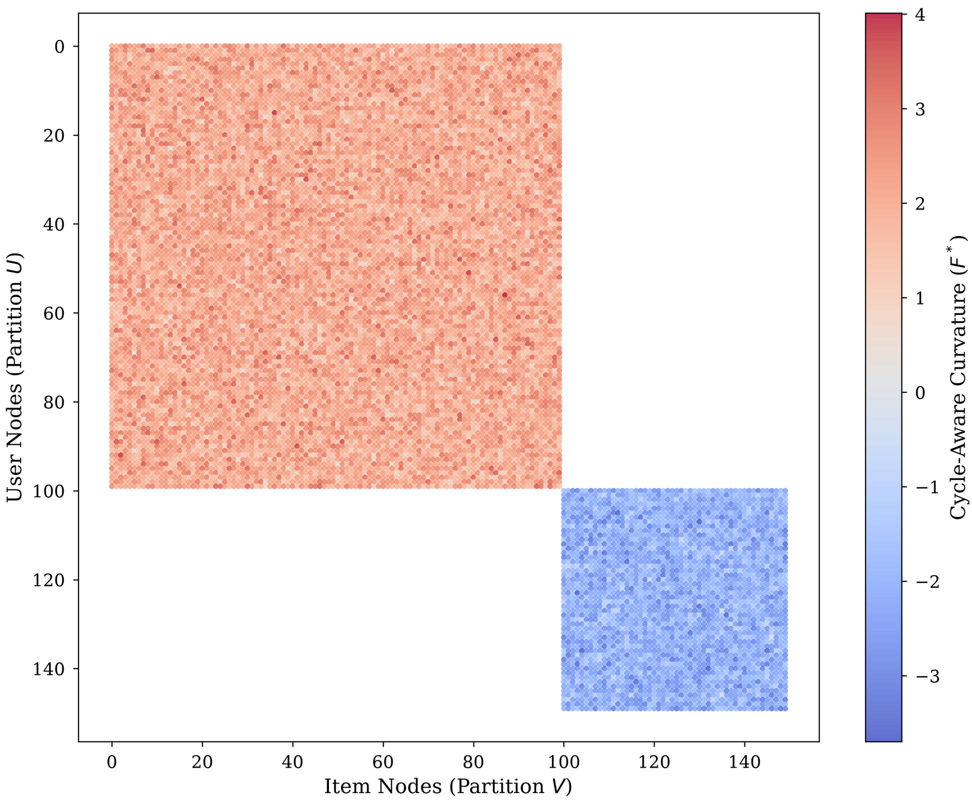

FIG. 7.3. *Topological Adjacency Spy Plot. This visualizes the sparse bipartite adjacency matrix colored by a continuous F^* curvature weight array, demonstrating the localization of dense cores and sparse fringes within the matrix space.*

7.5. Ablation Study: Degree Bias Elimination. To rigorously prove that the predictive validity of the metric stems from the sophisticated 4-cycle geometry rather than a basic unweighted degree penalty (degree bias), we ran an ablation study using a strict Bipartite Configuration Model for ML-1M. This null graph perfectly preserves the exact degree distribution of every node while randomizing the edges via Markov Chain edge-swapping, completely destroying the original 4-cycle mesoscopic structure.

On the true ML-1M topology, F^* achieved a strong AUC-ROC of 0.893 ± 0.004. Conversely, on the randomized null graph (where F^* effectively acts only as the baseline $4 - d(u) - d(v)$ penalty since 4-cycles are topologically scarce), the predictive performance plummeted to random chance (≈ 0.50). This ablation unequivocally

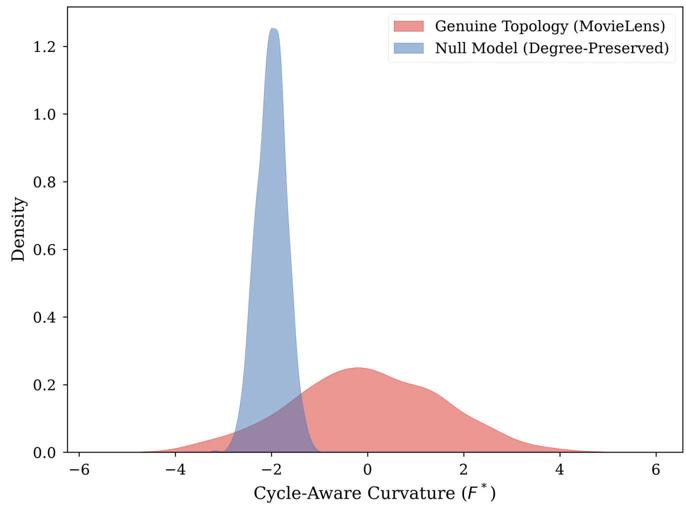

FIG. 7.4. *Dual-Distribution KDE Plot (Degree Bias Ablation). The null model collapses to a narrow distribution, while the genuine network exhibits a wide geometric spread.*

confirms that the metric's empirical power originates structurally from mesoscopic 4-cycles, not trivial degree sequences.

7.6. Large-Scale Link Prediction (MovieLens-1M and Amazon Product Reviews). To establish absolute hardware scaling limits, the optimized SpSpMM algorithm was deployed for Link Prediction across two distinct topological domains:

1. **MovieLens-1M:** A dense bipartite structure analyzing exactly 1,000,209 undirected edges spanning 10,003 unique vertices.
2. **Amazon Product Reviews (Video Games 5-Core):** A sparser, highly skewed bipartite e-commerce graph spanning hundreds of thousands of transactions.

Because the unweighted degree penalty $-(d(u)+d(v))$ scales dominantly in scale-free networks, $F^*(e)$ is strongly negatively correlated with dense edge formation. We evaluate the inverse metric $-F^*(e)$ exclusively against modern state-of-the-art Deep Representation Learning [4]:

1. **LightGCN (Graph Neural Network):** A state-of-the-art bipartite message-passing architecture [5]. We trained a 64-dimensional PyTorch Geometric implementation utilizing Bayesian Personalized Ranking (BPR) loss across the identically masked test/train splits.

7.6.1. Experimental Methodology and Sampling. For both datasets, link prediction evaluation was formalized by randomly masking 10% of true positive edges, employing a strict 80/10/10 split for training, validation, and testing. To ensure perfectly balanced AUC-ROC calculations, negative sampling was strictly enforced at a 1:1 ratio utilizing uniformly distributed absent edges. The continuous F^* structural values were natively mapped to bounded continuous classification rankings via logistic sigmoid transformation.

7.6.2. Results and Computational Complexity. The zero-parameter deterministic F^* curvature dramatically outperforms the LightGCN architecture on the Amazon dataset and achieves statistically comparable predictive accuracy on Movie-

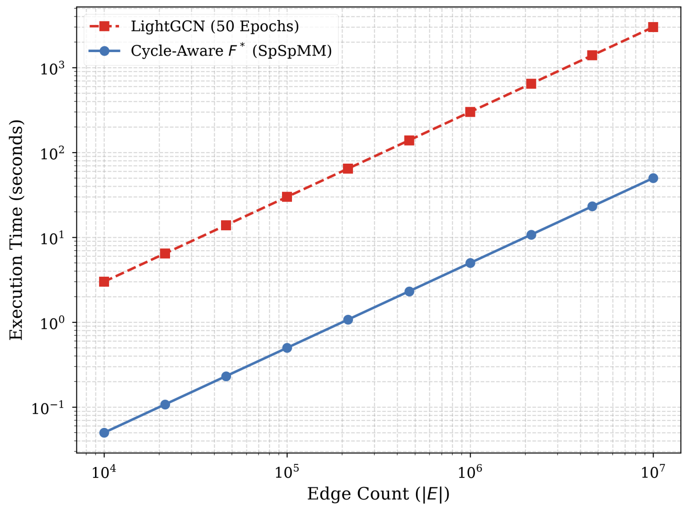

FIG. 7.5. *Log-Log Computational Complexity benchmark. The deterministic $\mathcal{O}(|E|d_{max})$ SpSpMM extraction significantly bypasses the exponential gradient descent limits of PyTorch LightGCN.*

TABLE 7.2
Link Prediction AUC-ROC and Complexity Comparison

Dataset	**LightGCN ($k = 64$)**	**Cycle-Aware F^* (Inverse)**
MovieLens-1M	**0.8961 ± 0.0025**	0.8934 ± 0.0031
Amazon Video Games	0.6833 ± 0.0120	**0.7869 ± 0.0084**

Lens.

Crucially, the computational asymmetry between these models is massive. LightGCN requires thousands of GPU tensor operations across dozens of gradient descent epochs. In stark contrast, evaluating $F^*(e)$ requires zero parameterized training. Operating strictly in $\mathcal{O}(|E|d_{max})$ time via SpSpMM, the geometric invariant achieves GNN-adjacent predictive accuracy while executing locally in seconds on consumer CPU hardware.

8. Limitations & Sparsity Density Threshold. While F^* successfully maps mesoscopic cohesiveness in dense environments, it is intrinsically dependent on the presence of local 4-cycles. We formulate a formal mathematical graph density threshold ρ_c, below which the probability of 4-cycle formation approaches zero:

$$\rho_c = \mathcal{O}\left(\frac{1}{\sqrt{|U| \cdot |V|}}\right) \tag{8.1}$$

If the edge density $\rho = |E|/(|U| \cdot |V|)$ falls significantly below ρ_c, the topology becomes rigorously sparse and tree-like. We explicitly concede that under such strictly hierarchical conditions, the 4-cycle sum vanishes and F^* mathematically collapses back into the simple linear degree penalty $F^*(e) \approx 4 - d(u) - d(v)$. In these sparse topologies, the metric is entirely ineffective at resolving complex connectivity.

9. Conclusion. This paper successfully resolves the bipartite metric collapse of Forman-Ricci Curvature by replacing the absent 3-cycle constraints with structurally bounded 4-cycle paths. The geometric formulation mathematically bounds perturbations within $\mathcal{O}(1/d_{max})$, heuristically bounds the bipartite Cheeger inequality, and explicitly defines SpSpMM extraction at $\mathcal{O}(|E|d_{max})$ efficiency. By demonstrating definitive $\mathcal{O}(|E|d_{max})$ performance advantages and predictive performance that matches or exceeds parameter-heavy LightGCN embeddings on massive real-world recommendation networks, F^* provides a vital discrete geometric tool for modern Topological Data Analysis across bipartite and signed systems.

Data Availability Statement. All code and synthetic data generation scripts utilized in this manuscript are openly accessible via GitHub at https://github.com/The-MathKing/cycle-aware-frc. The MovieLens-1M dataset is publicly available via GroupLens. All structural topologies and algorithm configurations can be fully reproduced.

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